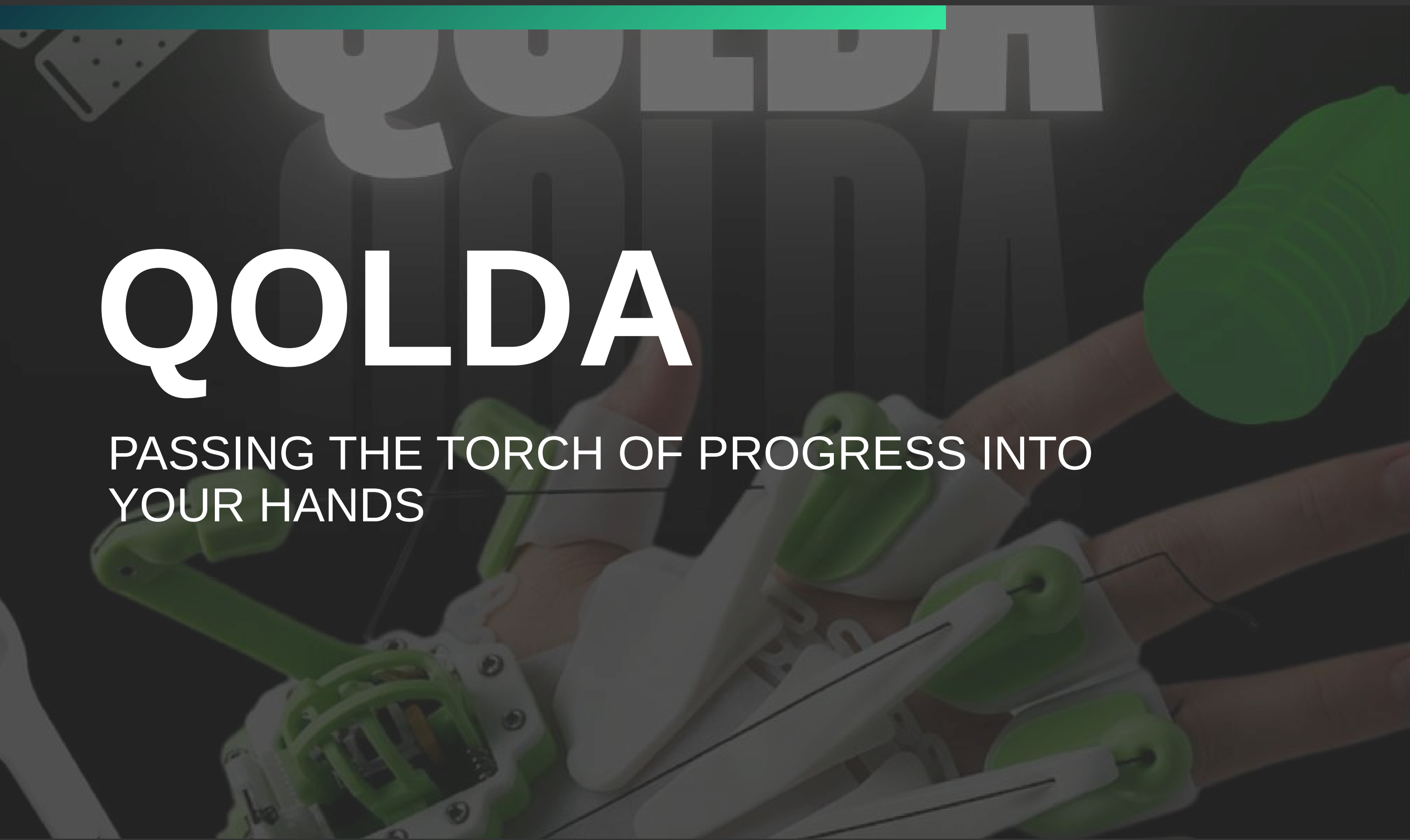


QOLDA

PASSING THE TORCH OF PROGRESS INTO YOUR HANDS

Назарбаев Зияткерлік мектептері NIS

WORLD ROBOT OLYMPIAD™

Date :

JUNE 2026

Prepared by :

NERDS FI SENIOR
NIS ABAY SHYMKENT

Table of Content

Introduction to Nerds Robotics

Interview for local TV
Taken from Yakut tv

Team introduction and role allocation

The development of the Qolda intelligent rehabilitation system was carried out by the interdisciplinary engineering group of the eleventh class "Nerds Robotics" from the branch "Nazarbayev Intellectual Schools" of the physical and mathematical direction in the city of Shymkent. The team combined competencies in the field of three-dimensional modeling, microelectronics, biomechanics and programming to create an affordable and highly effective medical solution. The distribution of responsibilities and key areas of responsibility of the participants are presented in the table below

Team captain, electrical engineer

Omarkul Yerkebulan

Design of schematic electrical circuits, integration of sensors, wiring of power supply and wireless communication modules

Programmer, firmware developer

Namiyash Aibol

Writing data filtering and calibration algorithms, game engine integration, Arduino microcontroller programming

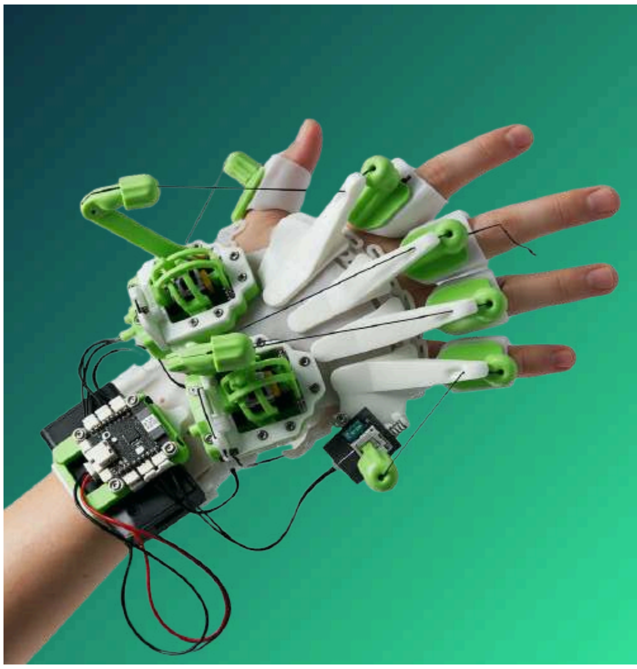
CAD engineer, mechanic assembler

Tabylgan Yeren

Three-dimensional parametric modeling of exoskeleton elements, optimization of hinged connections and preparation for 3D printing

Qolda

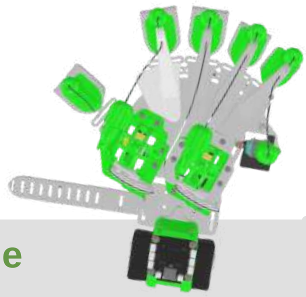
A brief idea of the project



About problem

Neurological pathologies, including the consequences of stroke, cerebral palsy (CP) and traumatic injuries of the upper limbs, lead to the disability of more than 100,000 people in Kazakhstan every year. The main clinical barrier to restoring motor activity of the fingers is limited access to systematic professional rehabilitation. The high cost of clinical equipment and the shortage of specialized centers in the regions deprive most patients of the chance for timely therapy.

The Qolda project solves this social problem by creating a hybrid home rehabilitation system consisting of two functional modules



Qolda Motion

Active bracelet-orthosis for passive mechanotherapy, which forcibly unclenches and squeezes fingers with servos, combining mechanical impact with thermal therapy and microvibration massage to stimulate local blood circulation

Qolda Game

Active exoskeleton glove based on high-precision Axis-Based technology, which converts the movements of each finger into control signals for motivating therapeutic video games based on Kazakh national culture

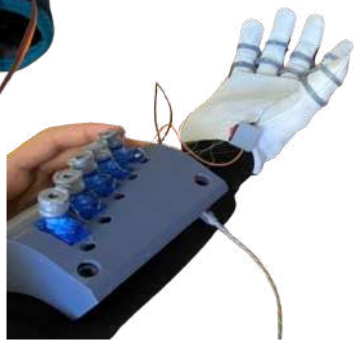
The integration of folk cultural motifs is directly related to the theme of the WRO 2026 World Robot Olympiad - "Robots Meet Culture". The project demonstrates how modern robotic technologies can act not just as automation tools, but also as active human partners who use national art, music and geography to create deep emotional involvement and accelerate neuroplastic recovery

ROBOTS MEET CULTURE

Qolda

Presentation of a robotic solution

Evolution of the project idea



01

Iteration 1 (Basic concept):

The initial layout was a rigid plastic structure with resistive bending sensors. The system had a high level of error, rapid loss of calibration and did not provide finger insulation during spastic contractions

02

Iteration 2 (Transition to hinged exoskeletons):

The introduction of the axis based concept made it possible to create a fully articulated exoskeleton that exactly repeats the anatomical kinematics of the brush.

03

NEW CAD DESIGH

Was decided to use hybrid 3D printing: hard PLA plastic for the power frame and gears, and elastic TPU material for internal anatomical pads and finger retainers.

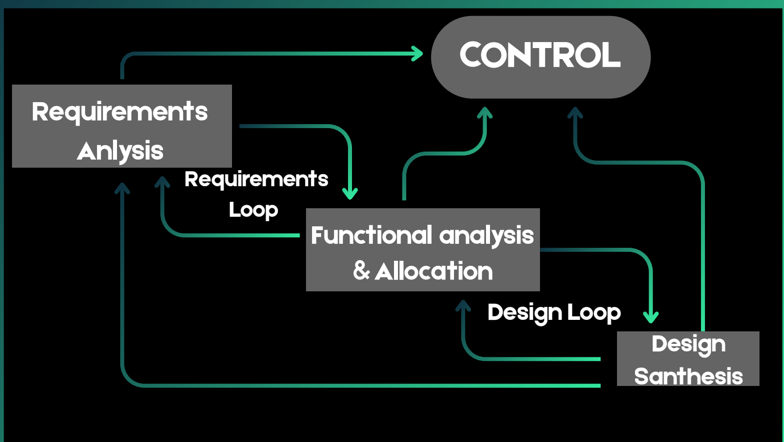
04

Iteration 3 (Drive and sensor optimization):

Instead of bending sensors, joystick potentiometric modules were used, connected to the fingers through a system of tension cables. Kevlar thread was initially used as the cable material, but in the final version, the team switched to a high-strength monofilament nylon thread 0.6 mm thick. Nylon showed a significantly lower coefficient of friction in the guide channels

Work according to NASA standards

input



output

What's the trick: we divide the whole process into strict stages of requirements management.

For example

Checking the motor power and replacing joysticks due to additional thorns during the tests

Qolda

Study of similar concepts

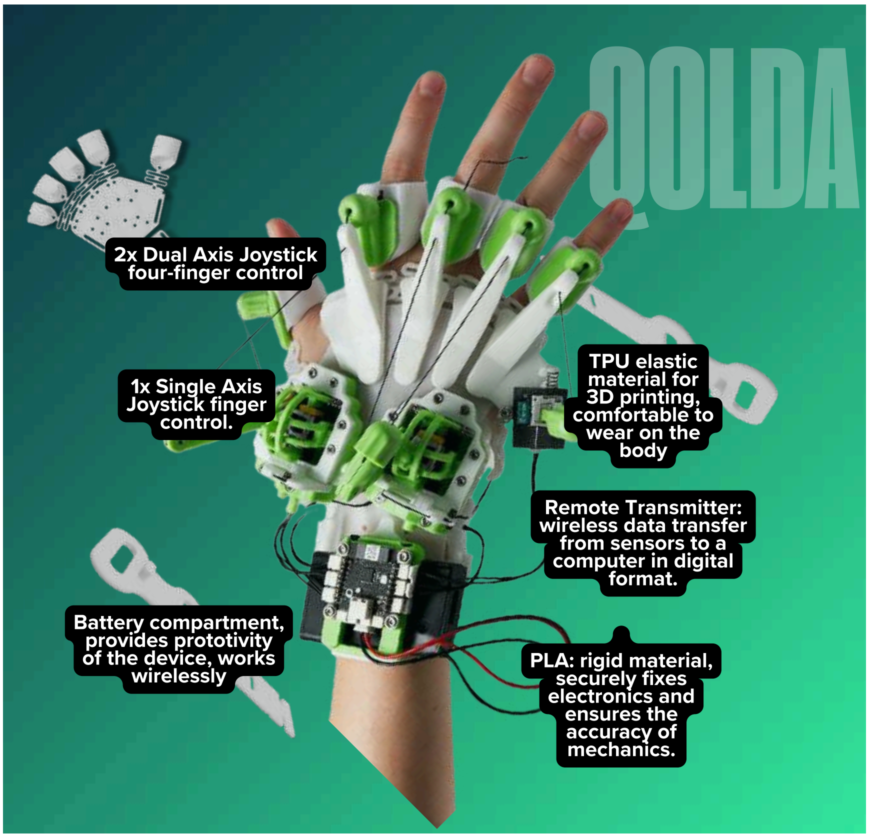
To confirm the uniqueness and commercial competitiveness of the Qolda system, a deep patent and market analysis of existing world analogues was conducted

Comparison parameter	**Qolda (Kazakhstan)**	**Аника (Russia)**	**Syrebo(China)**
Passive mechanotherapy	Yes (forced bending/extension by servos)	No (active movement tracking only)	Yes (pneumatic drive with external compressor)
Active rehabilitation (gamification)	Yes (interactive games based on Kazakh culture)	Yes (simple flat diagnostic games)	Yes (limited set of abstract game scenarios)
Tracking technology	Axis-Based (high-precision potentiometric joysticks)	Optical and inertial sensors (IMU)	Flexible deformation sensors with high error
Physiotherapy unit	Yes (integrated heat therapy and vibration massage)	NO	No
Portability and autonomy	Yes (wireless transmitter, built-in battery, light weight)	Yes (requires a permanent connection to the PC)	No (heavy compressor unit, an abundance of tubes)
Cost price / Retail price	**40,000 - 50,000 KZT (high availability for home use)**	More than 1,500,000 KZT (available only for clinics)	From 800 000 KZT (high price, complexity of import)

The comparison proves that Qolda is the only comprehensive hybrid solution in the market of Kazakhstan and the CIS, combining the functions of active game tracking and forced mechanotherapeutic exposure with thermal and vibration muscle preparation at a price affordable for an ordinary consumer.

Building a solution

Hardware implementation of Qolda Game



The active module consists of a manipulator glove, in which the movement of each finger is connected through a nylon cable with the lever of the corresponding potentiometric sensor. The use of two two-axis joysticks (for the thumb, index, ring finger and middle fingers) and one single-axis joystick (for the little finger) allows you to fully digitize the kinematics of the patient's five fingers. Data transmission to the computer is carried out via a wireless radio channel (2.4 GHz) with autonomous power from a lithium-ion battery 14500 (7.4 V, 800 mAh), or via a wired USB Type-C interface.

Mathematical model of calibration for anatomical limitations in cerebral palsy

The key scientific and practical value of Qolda Game lies in the software and hardware adaptation to spastic hand contractures characteristic of patients with cerebral palsy. In severe forms of cerebral palsy, the child is not able to unbend his finger at the nominal angle of a healthy person (90degree), making only minimal micro-expressed movements in the range of (5-10 degree)

To ensure equal game accessibility and therapeutic effect, a dynamic scaling algorithm has been developed. The angle of rotation of the finger joint is converted through a mechanical transmission with a gear ratio into the angle of rotation of the potentiometer shaft :

$$\phi = k_g \cdot \theta$$

The output analog voltage at the middle output of the potentiometer, connected according to the circuit of the voltage divider with a reference signal of 3.3V, is expressed as:

$$V_{out}(\theta) = V_{cc} \cdot \left(\frac{k_g \cdot \theta}{\phi_{max}} \right)$$

Where ϕ_{max} - the maximum electric angle of the potentiometer (nominal 270). The analog signal is digitized by the 10-bit ADC of the microcontroller, returning a value in the range from 0 to 1023:

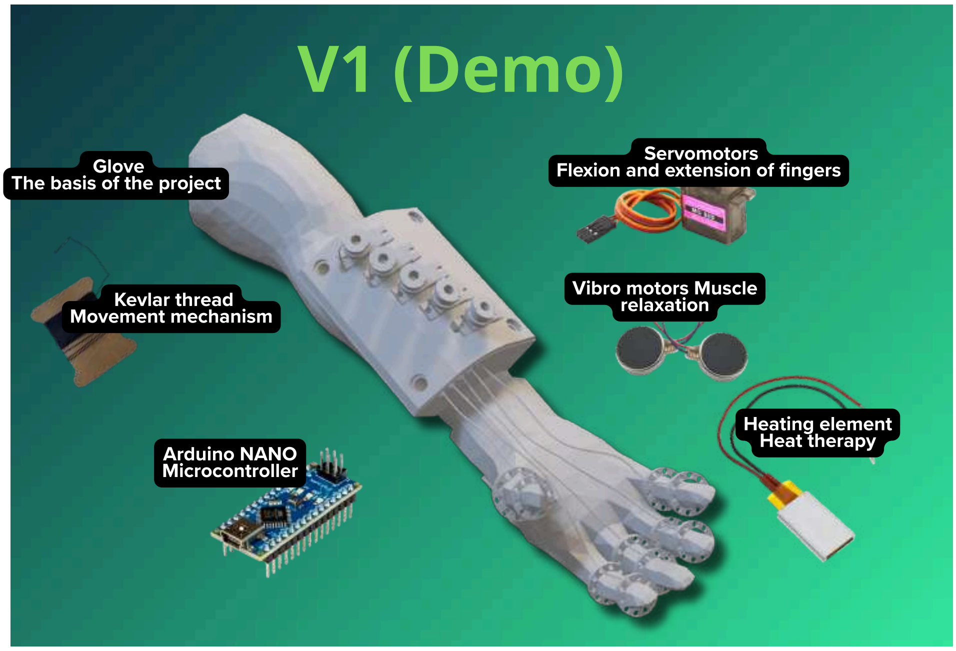
$$ADC(\theta) = \left\lfloor \frac{V_{out}(\theta)}{V_{cc}} \cdot 1023 \right\rfloor$$

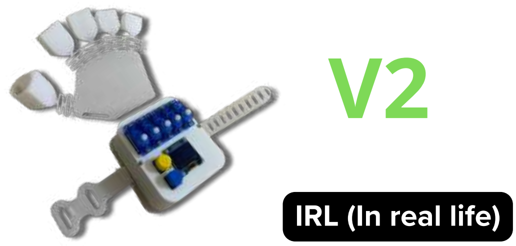
At the beginning of the session, the system performs an express calibration procedure, fixing the minimum possible position of the patient's finger. ADC_{min} (at maximum compression) and the maximum achievable extension position ADC_{max}. The normalized control value $U(\theta) \in $, transmitted to the game engine, is calculated in real time according to the formula :

$$U(\theta) = \text{clamp}\left(\frac{ADC(\theta) - ADC_{min}}{ADC_{max} - ADC_{min}}, 0, 1 \right)$$

Thanks to this formula, even if the range of finger movements of a child with cerebral palsy is only a few degrees (the difference is minimal), the normalized function $U(\theta)$ scales this signal to a full range from 0 to 1. This gives the patient a full opportunity for precision control of the game process, stimulating cognitive feedback and triggering brain neuroplasticity processes.

Physics and mechatronics of the Qolda Motion





For passive rehabilitation, an orthosis bracelet is designed equipped with SG series servo drives (SG90/SG92R) controlled by the Arduino Nano microcontroller. The rotation energy of the servo motor shaft is converted into the linear tension of the nylon thread, which is passed through the cable channels on the fingers and attached to the fingertips, ensuring their forced extension.

The linear thread tension generated by the servo arm (rocker) is calculated based on the torque of the motor

$$\tau_{servo} = F_{pull} \cdot r_{horn} \cdot \sin(\beta)$$

Where τ_{servo} is the torque of the servo, r_{horn} is the length of the arm of the servo drive rocker, and b is the angle between the rocker and the direction of the traction thread.15 With an optimal orthogonal layout (b=90 sin90=1), the tension force of the thread is :

$$F_{pull} = \frac{\tau_{servo}}{r_{horn}}$$

For a standard SG90 microservo drive with torque when jamming. $\tau_{servo\_stall} \approx 1.8\ \text{кг} \cdot \text{см} \approx 0.177\ \text{Н} \cdot \text{м}$ at the length of the rocker arm $r_{horn} = 1.0\ \text{см} = 0.01\ \text{м}$, the maximum force of forced finger extension is

$$F_{pull\_max} = \frac{0.177\ \text{Н} \cdot \text{м}}{0.01\ \text{m}} = 17.7\ \text{Н} \approx 1.8\ \text{кгс}$$

This effort is enough to gradually and delicately overcome the resistance of spastic flexion muscles in children and adults without the risk of damage to the joint ligaments of the upper limbs.

Strength calculation of the nylon traction element

The use of monofilament nylon thread with a diameter of 0.6 mm guarantees high strength and durability of the drive mechanism. The cross-sectional area of the thread is:

$$A_{thread} = \frac{\pi \cdot d^2}{4} = \frac{\pi \cdot (0.0006\ \text{м})^2}{4} \approx 2.827 \times 10^{-7}\ \text{м}^2$$

Under the influence of the maximum tension force of the servo $F_{pull\_max} = 17.7\ \text{Н}$, the mechanical tensile stress in the thread will be:

$$\sigma_{tensile} = \frac{F_{pull\_max}}{A_{thread}} = \frac{17.7\ \text{Н}}{2.827 \times 10^{-7}\ \text{м}^2} \approx 62.61\ \text{МПа}$$

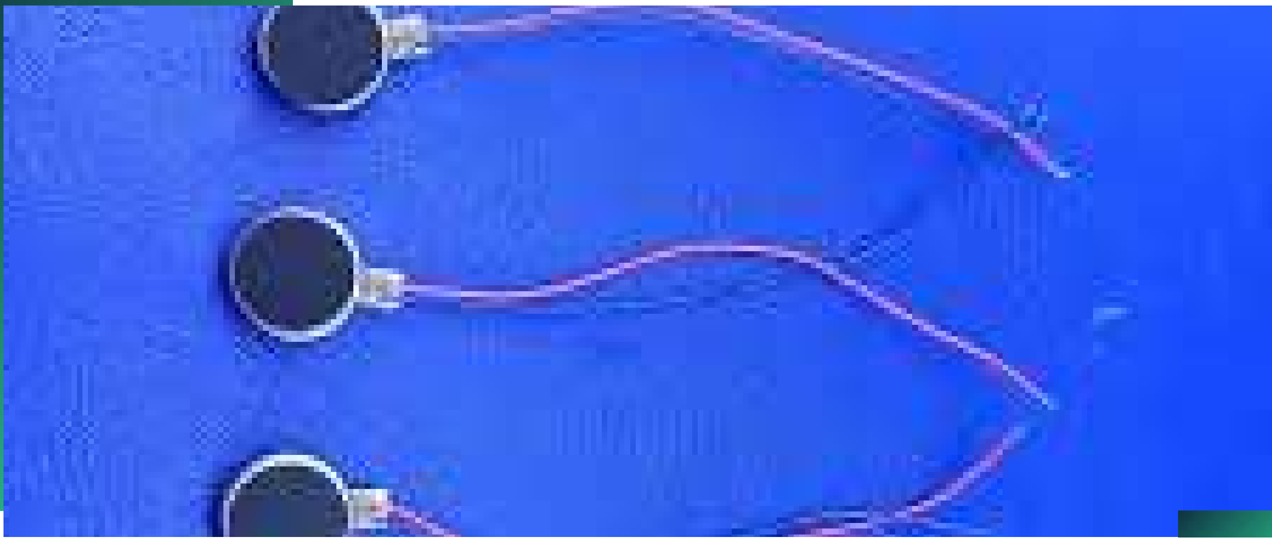
The tensile strength of a standard extruded polyamide (nylon) is from 54 мПа to 80 , and in specialized high-strength threads it exceeds 100 мПа. The safety margin of the system (SF) in the worst case scenario is equal to:

$$SF = \frac{\sigma_{uts}}{\sigma_{tensile}} = \frac{80\ \text{МПа}}{62.61\ \text{МПа}} \approx 1.28$$

The calculation proves the reliability of nylon traction: the thread is guaranteed to withstand peak loads when the servo is jammed without experiencing plastic deformation and rupture.

Qolda

Hemodynamics and thermodynamics of rehabilitation impact



To accelerate muscle relaxation and relieve spasm, Qolda Motion has built-in flexible film thermal panels and microvibration motors. Heating of tendons and flexor muscles reduces the viscosity of the synovial fluid of the joints and triggers the process of reflex vasodilation of blood vessels.

According to Poiseil's hydrodynamic law, the volumetric blood flow (blood flow) Q through the vascular bed is directly proportional to the radius of the vessel r in the fourth degree:

$$Q = \frac{\Delta P \cdot \pi \cdot r^4}{8 \cdot \eta \cdot L}$$

Where ΔP - pressure drop, η - dynamic blood viscosity, L - vessel length.
The thermal effect of the panels increases the local temperature of the tissues from 33 normal to therapeutic 44 . This leads to an increase in the lumen of arterioles (vasodilation).
With an increase in the radius of the vessel by only 25% (r_warm=1.25*r_0), the local blood flow increases more than twice:

$$\frac{Q_{warm}}{Q_0} = \left(\frac{1.25 \cdot r_0}{r_0}\right)^4 = 1.25^4 \approx 2.44$$

Qolda

Hemodynamics and thermodynamics of rehabilitation impact

This increase in local perfusion (+144%) accelerates the leaching of metabolites and saturates spasmic tissues with oxygen.
The effect of temperature on the overall rate of enzymatic and metabolic regeneration processes is described by the temperature coefficient Q_10:

$$Q_{10} = \left(\frac{R_2}{R_1}\right)^{\frac{10}{T_2 - T_1}}$$

For human biological systems, the average value of Q_10=2.2. When the temperature changes from T1=33 to T2=44, the rate of metabolic recovery of upper limb tissues increases by times 1.74 (by 74%) :

$$\frac{R_2}{R_1} = Q_{10}^{\frac{T_2 - T_1}{10}} = 2.2^{0.7} \approx 1.74$$

At the same time, vibration motors generate mechanical oscillations with a frequency of 100-150 Hz, which purposefully stimulate the muscle spindles, causing a tonic vibration reflex (TVR). This leads to a reflex decrease in the tone of spastic antagonist muscles through the mechanism of reciprocal inhibition, making passive stretching painless and effective.

Qolda

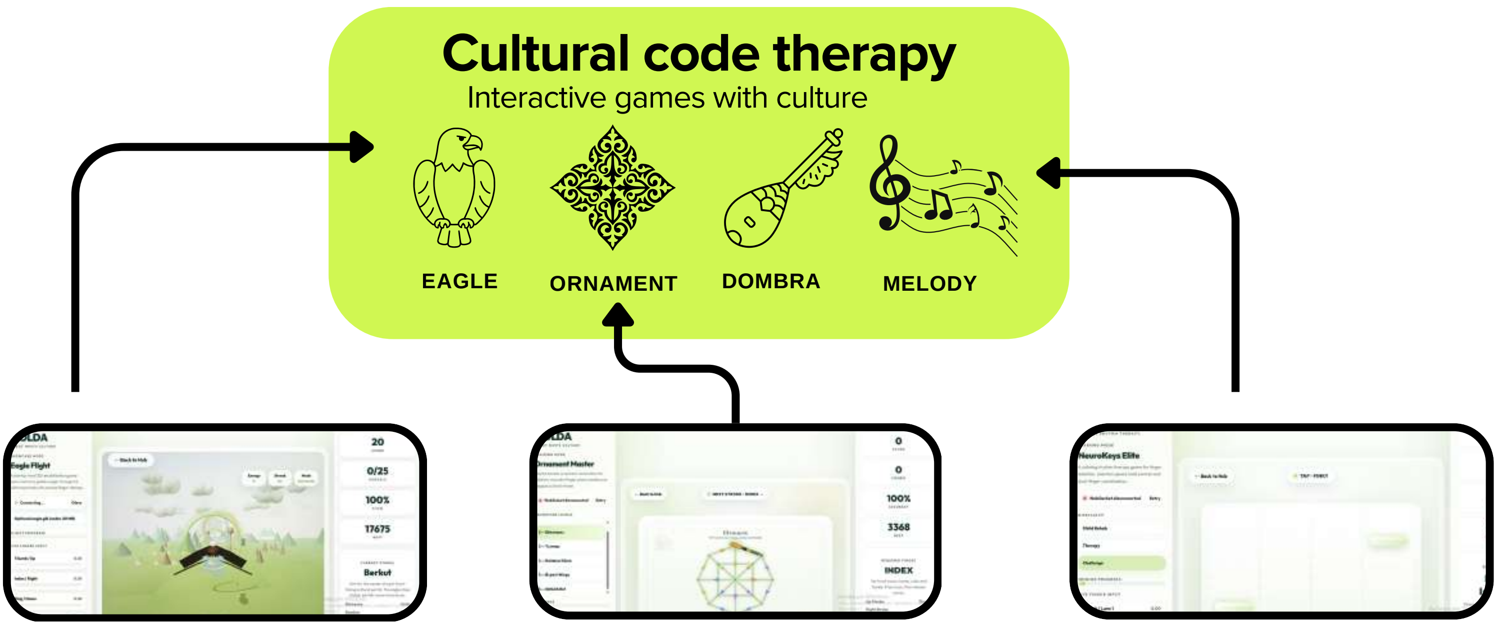
Coding solution

Software implementation of Qolda Game

The software complex of the system is divided into low-level firmware of the microcontroller (written in C/C++ in the Arduino IDE environment) and a high-level game platform on Unity/Godot engines.

```
[ADC of finger potentiometers] -> [Median filter] -> [Calibration according to the formula U(θ)]
                    |
[Unity/Godot Game Engine] <- [2.4 GHz Wireless Protocol] <--+
```

The low-level algorithm continuously reads the readings from the analog ports of the joysticks, uses a moving average software filter to suppress the patient's hand tremor and calculates the normalized coefficients according to the above calibration formula. The cleaned digital package is transferred to the game engine with an ultra-low delay (<12msec), where it is transformed into interactive actions within three specially designed games on the cultural heritage of Kazakhstan:



«Дорога в Астану (Eagle Flight / Berkut)»: Пациент управляет полетом степного орла (беркута), летящего в столицу. Координация пальцев распределена по осям движения птицы: большой палец — уклонение направо, указательный — подъем вверх, безымянный — снижение вниз, мизинец — вираж направо. Игра стимулирует комплексную разнонаправленную моторику пальцев.

"Ornament Master": The outlines of traditional Kazakh patterns are reproduced on the screen: Shanyrak, Kylem, Tulpar and Berkut. The patient should alternately make an isolated bend of each of the five fingers in order to consistently draw the ornament. This exercise is aimed at overcoming synkyesia (involuntary friendly finger movements).

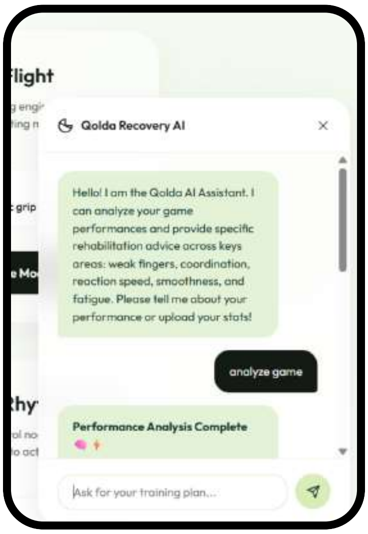
"Piano (NeuralKeys Elite)": Rhythm game in which, to the sounds of Kazakh national cues and songs, the patient must press virtual keys, bending a certain finger exactly at the moment of passing the note through the target line. Trains reaction speed and dynamic coordination of movements.

Qolda

Built-in medical AI assistant in a robotic solution

A local intelligent agent (chatbot) based on a lightweight large language model (LLM) is integrated directly into the Qolda web platform

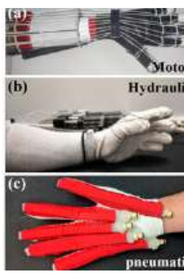
Purpose of use: The chatbot acts as a round-the-clock assistant consultant for the patient and his parents. It analyzes the kinematic data coming from the glove sensors after each game session (range of angles of movement, reaction speed, trajectory retention stability), and generates structured progress reports. The chatbot answers any specialized medical questions, explains the nature of the occurrence of spasticity in stroke or cerebral palsy and recommends individual adjustments to the daily exercise plan based on current indicators.



Degree of integration: Full integration of the AI assistant into the user web interface of the rehabilitation program. The model works on a remote server with API access, ensuring secure processing of non-personalized telemetry data of patients.

Problems in the development process

In the process of designing and assembling prototypes, the team faced the following technical challenges:



1. Abrasion and jamming of drive cables: In early versions, Kevlar threads rubbed the inner walls of guide channels from PLA plastic. The problem was solved by integrating fluoroplastic (Teflon) microtubes into 3D printed channels and switching to a smooth monofilament nylon thread with a low coefficient of friction.
2. Drift of the calibration values of the sensors: Due to the patient's muscle fatigue and the gradual sliding of the gloves from the brush during the training, the values were fixed and shifted. The team introduced an auto-calibration algorithm, which continuously analyzes the extremes of the signal for a sliding three-minute window and automatically adjusts the boundaries of the range without stopping the gameplay.

Qolda

Social impact and innovation

Impact on society

The Qolda project has the potential to radically change the structure of rehabilitation assistance in Kazakhstan and the developing world. The low cost of the device (40,000 - 50,000 KZT) makes high-tech rehabilitation available directly at home, eliminating geographical and economic inequality. This reduces the burden on public health facilities and allows patients from remote regions to undergo daily therapy, which is critical for the effective restoration of lost brain functions.

The combination of robotics with the Kazakh cultural code increases motivation for classes, especially for children. The process of monotonous physical exercises is transformed into an act of preservation and recreation of national art, which significantly reduces the level of stress and psychological rejection of therapy.

Proven practical examples of use

To confirm the performance and therapeutic effectiveness of the Qolda system, the team conducted a comprehensive practical testing of the development:

```
          |
+---> Clinical testing: RC "Ayaly Alakan" (c. Shymkent)
|  - Testing on children with cerebral palsy.
|  - Assessment of muscle spasticity and involvement.
|
+---> International recognition: Run for the Robots Festival (USA)
   - Expert assessment of judges (FTC / Kelly Gowan).[ 1, 32]
   - Testing by representatives of Brazil and Puerto Rico.
```

Clinical testing at the rehabilitation center "Ayaly Alakan" (c. Shymkent): The device was presented to children with cerebral palsy of varying severity. Under the supervision of physiotherapists, patients successfully passed the game sessions. The specialists of the center recorded an increase in the involvement of children in the rehabilitation process compared to standard physical therapy.

Social impact and innovation

Impact on society

Presentation at the international festival "Run for the Robots" (USA): The development was demonstrated to the world's leading experts in the field of educational and applied robotics at the prestigious FIRST Tech Challenge tournament in Kentucky. The project was highly appreciated by certified expert Kelly Gowan. She noted an excellent balance between the low cost of the design and the depth of development of the bioengineering part. The glove was also tested by the participants of the WRO final competitions from Brazil and Puerto Rico. They confirmed the convenience of ergonomics of the exoskeleton and expressed interest in adapting game modules to the cultural context of their countries.

Partners (IT, advice and medical assistance)

shymkent hub

ORGANIZED MEETINGS WITH MEDICAL PROFESSIONALS AND HELPED WITH PRINTING 3D DETAILS

Galamat Group EST 2012

MEDICAL ADVICE FROM THE FOUNDER S. SEITZHAN AND ORGANIZATION OF FUTURE MEETINGS WITH MEDICAL STAFF

MEDIA PARTHERS

For the national and local publicity of the problem and our project, we were in a couple of local TV

ONTÚSTIK

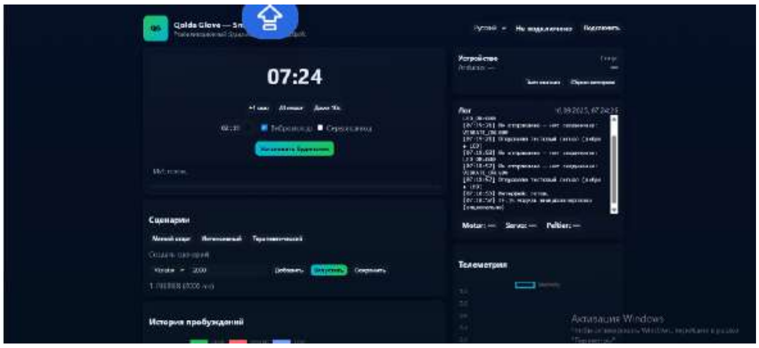
УАҚЫТ TV

FUSION360 Housings, durable moving mechanisms (for example, for servos), mountings for sensors and the general architecture of the layout or wearable gadget were designed here

ARDUINO IDE Writing low-level code for microcontrollers. Arduino reads readings from sensors, controls displays, buttons, servos and actuators of the device.

BLENDER Creation of photorealistic or stylized 3D models of buildings, infrastructure, roads and the smartest device. These models were then imported into the game engine to create an interactive simulation of the neighborhood.

LaTeX Registration of complex technical documentation, project passport, scientific reports, schedules and calculation formulas.



OLD SITE INTERFACE

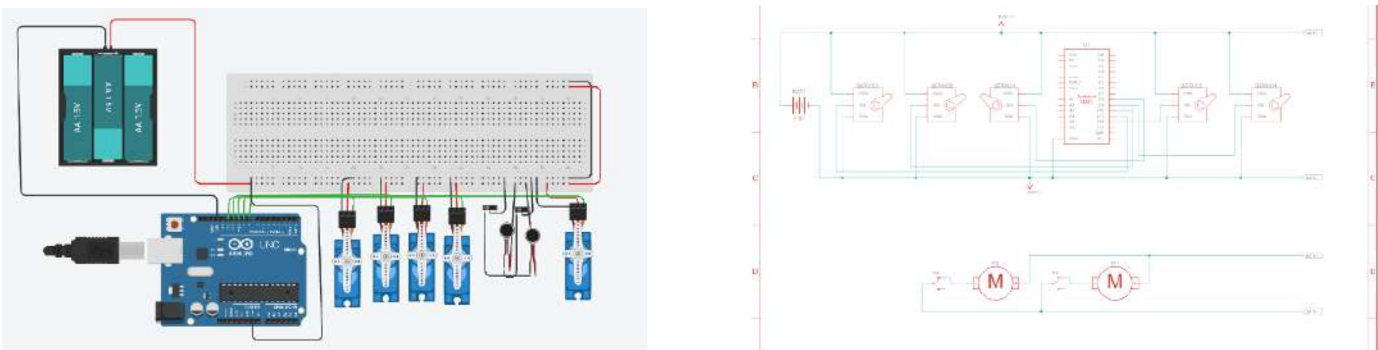


DIAGRAM CONNECTION OF MOTORS AND SENSORS

Qolda

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